Can LLMs Provide Objective Knowledge Bases for Math and Physics, or Does ASI Require a Specialized Format?

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Executive Summary

For an Artificial Superintelligence (ASI) limited to mathematics and physics, Large Language Models (LLMs) alone cannot constitute an objective knowledge base. The research consensus from 2023-2026 converges on a single conclusion: LLMs are probabilistic continuation engines whose training objective rewards plausibility, not truth. Hallucination is not a bug but a structural property, and recent mathematical work shows it is inevitable.

OpenAI's own 2024-2025 research acknowledges that hallucinations persist due to three irreducible factors: epistemic uncertainty on long-tail facts, representational limits, and computational intractability, with reasoning models hallucinating more (o1: 16%, o3: 33%, o4-mini: 48% on summarization tasks). In mathematics and physics specifically, LLMs fail fine-grained derivation: they reverse-engineer solutions, ignore physical semantics (units, tensor order, conservation laws), and produce algebraically coherent but physically impossible reasoning.

ASI in this domain therefore requires specialized, verifiable formats:

1. Formal proof assistants (Lean 4, Coq/Rocq, Isabelle) with libraries like mathlib (>210,000 theorems as of May 2025) and PhysLib/PhysLean (180k+ LOC) providing machine-checked truth.
2. Symbolic computation and SMT solvers (Wolfram|Alpha, SymPy, Z3) as deterministic calculation gates.
3. Explicit knowledge graphs and ontologies (weighted physics equation graphs, physmath-knowledge-tree) for provenance, traceability, and updateability.
4. Neuro-symbolic architectures where LLMs act as proposers and formal systems act as verifiers (DeepMind AlphaProof/AlphaGeometry 2: silver-medal IMO 2024 with Lean-checked proofs; Pythos: 0% incorrect delivery via CAS gate).

The answer is therefore not either/or but layered: LLMs provide intuition, paraphrase, and auto-formalization, while the objective knowledge base must be external, formal, and mechanically verifiable. For math/physics ASI, the specialized format is mandatory; the LLM alone is insufficient.

1. Introduction: Domain-Specific Superintelligence

Artificial Superintelligence is typically defined as intellect greatly exceeding human performance in virtually all domains. A narrowed ASI limited to mathematics and physics is both more tractable and more demanding for truth: unlike open-domain chat, mathematical claims are falsifiable by proof, and physical claims by dimensional analysis, conservation laws, and experiment. This creates a unique requirement for an objective, updatable, provenance-aware knowledge base.

The literature on narrow-domain superintelligence distinguishes systems like chess engines, AlphaFold, and recent physics reasoning systems that exceed general LLMs in a verified subdomain while being structurally bounded against hallucination. For math/physics, the bound must be formal verification.

2. LLMs as Implicit Knowledge Bases: Structural Limits

LLMs store knowledge implicitly in weights. Research by Petroni et al. (2019) and Lewis et al. (2020) established the distinction between parametric memory (internal, opaque, frozen at training time, unable to cite) and explicit memory (external, revisable, traceable). Philippe Buschini (2025) summarizes: 'An LLM is not a knowledge base. It excels at unrolling coherent sentences... But it verifies nothing, cites nothing, doesn't update itself.'

Key structural reasons for hallucination, per the deep-dive taxonomy (Maynez et al. 2020 and ffaisal93):

- • Training objective: next-token prediction rewards plausible continuation, not truth. Plausible ≠ true.
  • Coverage gaps and long-tail forgetting: power-law distribution means rare theorems, constants, or proofs are barely represented.
  • Tokenization quirks: numbers and DOIs split inconsistently, preventing clean memorization.
  • Position bias / lost in the middle: attention concentrates on start/end, mid-context facts are filled in.
  • RLHF reward hacking: human raters prefer confident, fluent answers, so models learn never to say 'I don't know'; calibration worsens after RLHF.
  • Compounding errors: chain-of-thought multiplies per-step error; 20 steps at 97% per step ≈ 54% overall.

OpenAI researchers (Kalai, Zhang, Nachum, Vempala, 2024-2025) formalized lower bounds: generative error rate is at least twice the Is-It-Valid misclassification rate, and hallucinations are mathematically inevitable due to epistemic uncertainty, model capacity limits, and cryptographic intractability. Their own evaluation showed GPT-5 still hallucinates, and reasoning models hallucinate more often despite longer chains.

3. Empirical Evidence of Failure in Mathematics and Physics

Fine-grained physics derivation studies (Meadows et al., arXiv:2404.18384) curated 218 step-wise derivations across electromagnetism (82), quantum mechanics (98), and classical/statistical mechanics (38). While GPT-3.5, GPT-4 and T5 scored highly on automatic text metrics, manual evaluation found incoherent physics: violations of laws and assumptions. Under premise-removal interventions, quality degraded non-linearly, suggesting reverse-engineering of solutions rather than principled reasoning.

The Pythos project documents the same pattern in education: LLMs drop negative signs, fabricate intermediate arithmetic, and deliver confidently incorrect 6-step calculus integrations, derailing student understanding. The developers concluded 'never let the language model deliver unchecked math' and introduced a deterministic CAS gate that validates each algebraic step and withholds unverified answers, achieving 0% incorrect delivery.

Benchmark saturation reveals the ceiling:
• GSM8K and MATH: >90% for frontier models, now saturated.
• MMLU College Physics: ~91% → 95% with reasoning, saturated.
• GPQA Diamond (Google-proof graduate science): best ~55-60% (Qwen, 2026), hard because it requires multi-concept reasoning.
• FrontierMath (Epoch AI, several hundred unpublished research-level problems): GPT-4o and Gemini 1.5 Pro <2% in 2024; o3 reached 25.2% (Tier 1-4) and GPT-5.5 51.7% Tier 1-3 / 35.4% Tier 4 in 2026. Small capability drops cause massive score drops, indicating lack of robust long-chain logic.

4. What Objective Knowledge Means for Math and Physics

For ASI limited to these domains, objective means:
• Mechanically verifiable: a proof term checked by a small trusted kernel (Lean's kernel) yields binary correct/incorrect, not a probability.
• Provenance-aware: each fact has URI, author, date, source, and dependency chain.
• Updatable without retraining: new theorems, constants, or experimental values added to a library, not to weights.
• Dimensionally and logically consistent: units, tensor order, conservation laws enforced by type system.
• Refusable: system can output 'cannot prove' rather than guessing, avoiding reward for guessing.

The unification roadmap (Pan et al., arXiv:2306.08302) defines three patterns: KG-enhanced LLMs, LLM-augmented KGs, and synergized LLMs+KGs. For objective domains, the third pattern dominates: LLMs and KGs play equal roles, with KGs providing external knowledge for inference and interpretability, and LLMs helping construct and query the KG.

5. Specialized Formats Required for ASI

5.1 Formal Proof Assistants: Lean 4, Mathlib, PhysLib

Lean 4 is a functional programming language and proof assistant based on the Calculus of Inductive Constructions. Every theorem must pass strict type-checking by Lean's trusted kernel. Advantages: precision, systematic verification, transparency, reproducibility. Mathlib (as of May 2025) formalized over 210,000 theorems and 100,000 definitions. PhysLib (formerly PhysLean + Lean-QuantumInfo, 180,646 LOC, Apache-2.0) aims to be the definitive physics library, with sectors for classical mechanics, EM, QFT, quantum information, relativity, thermodynamics. HepLean/PhysLib already found a 2006 error in Two-Higgs-Doublet Model literature via formalization.

Lean4Physics (Li et al., arXiv:2510.26094, ICLR 2026) extends this with LeanPhysBench: 200 hand-crafted college-level physics theorem statements formalized from textbooks and competitions. Key finding: expert provers overfit to math; with PhysLib in context, LLM performance on LeanPhysBench increases 11.90% on average. This demonstrates that a formal library is not optional decoration but directly improves reasoning.

5.2 Symbolic Computation Engines and Solvers

Wolfram|Alpha LLM API, SymPy, and SMT solvers like Z3 serve as deterministic verification gates. The pattern in Pythos: Student Input → LLM formulates steps → CAS Gate validates each algebraic operation → Audit → Verification Gate withholds if unverified. Wolfram provides exact arithmetic, symbolic algebra/calculus, unit conversions, physical constants, and structured data queries—eliminating numerical hallucination modes that are inherent to token prediction.

5.3 Knowledge Graphs and Ontologies

A 2025 graph-based framework (arXiv:2508.05724) built a corpus of 659 physics equations, cleaned to 400 advanced equations after resolving notational polysemy in 213 equations, represented as a weighted knowledge graph where nodes are concepts and edges are typed mechanisms (not mere 'related to'). Another project, physmath-knowledge-tree (Eriksson 2026), builds a computable graph with 59 domain/bridge/problem nodes and 115 typed, evidence-labelled edges (formal/literature/heuristic/speculative), with Lean target audit queues and bilingual prerequisite maps (90 topics, 199 edges). These provide the explicit, updatable memory that parametric models lack.

5.4 Physics-Informed Neural Networks and Verified Simulators

Physics-Informed Neural Networks (PINNs) embed physical laws (mass conservation, flow equations) directly into training objectives, penalizing physically impossible solutions (e.g., rain falling upward). Language models were shown to be geometrically blind on MatBench: graph neural networks (CogN, SchNet, DimeNet) clustered at low error, while language models (LLM-Prop, CrabNet) clustered at high error. For material property prediction, bigger was not better; architecture matters more than scale.

5.5 Neuro-Symbolic Architectures: AlphaProof, AlphaGeometry, and Beyond

The dominant recipe for verified math ASI is neuro-symbolic: LLM as creative proposer, formal system as verifier, synthetic data generated by verifier itself.

• AlphaProof (DeepMind, Nature 2025, s41586-025-09833-y): pretrained LM + AlphaZero-style RL over Lean 4 proofs. Generated ~80M auto-formalized statements from ~1M NL problems, MCTS with product nodes for multi-subgoal tactics, test-time RL on millions of variants. Combined with AlphaGeometry 2, solved 4/6 IMO 2024 problems (P1, P2, P6 by AlphaProof, P4 by AlphaGeometry 2, 28/42 points = silver-medal score, gold cutoff 29). Each AlphaProof solution is a Lean proof term checked by Lean's kernel—soundness does not rest on human judgment. Solve times ranged from 19 seconds (P4) to up to 3 days, and problem statements were human-formalized, revealing that 'silver medal' describes score, not contest-equivalent conditions.
• AlphaGeometry: symbolic engine + specialized LLM trained on synthetic data.
• Pythos, Principia AI, Skeptical Alpha, FormulaGate: all wrap deterministic math engines inside multi-agent loops, proving that generating truth requires external verification.
• Anthropic (Sept 2026): agents with internal model comparable to Claude Fable 5.1 produced complete machine-checked formalization of Fermat's Last Theorem in Lean 4 in eleven days (~13M lines, ~30,000 theorems), coordinated via Prove2Me platform, verified by Kevin Buzzard (Imperial). While mathematically adding nothing to Wiles, it demonstrates scalability of agentic formalization.

6. Comparative Analysis: LLM vs Specialized Format

Dimension	LLM as Knowledge Base	Specialized Format (Lean/Mathlib/PhysLib + KG + CAS)
Truth criterion	Probabilistic plausibility (next-token likelihood)	Mechanical verification (kernel type-check, CAS evaluation, KG provenance)
Updateability	Frozen at training; retraining costly, risks regression	Live: add/remove theorems, constants, edges without retraining
Provenance	Cannot cite; source hallucinated	Every fact has URI, author, date, evidence label
Rare facts / long tail	High hallucination on rare theorems	Exact lookup; frequency-independent
Dimensional consistency	Often ignored (units, tensor order)	Enforced by dependent types
Refusal behavior	Rewarded for guessing; RLHF worsens calibration	Explicit abstention gate; 0% incorrect delivery possible
Verification cost	Cheap inference, expensive trust (human review)	Expensive formalization upfront, cheap verification thereafter
Benchmark ceiling	<2% FrontierMath 2024,	100% soundness conditional

	~25% 2026 reasoning models	on correct formalization (Lean)

The table makes clear: LLMs and specialized formats are not competitors but complements. LLMs handle fuzzy formalization (auto-formalization of NL to Lean achieved for P1/P2/P6 in side experiment), intuition, and explanation; specialized formats handle objective truth. Confusing parametric memory for explicit memory, as Buschini warns, leads to governance failures (e.g., French Government video with hallucinated flags).

7. Benchmark and Evaluation Data

• Hallucination inevitability: OpenAI o1 16%, o3 33%, o4-mini 48% on summarization (2025 paper); generative error rate ≥ 2× IIV misclassification.
• Fine-grained physics: 218 derivations, automatic metrics high, manual physics coherence low; premise-removal shows non-linear degradation.
• FrontierMath: <2% (GPT-4o, Gemini 1.5 Pro, 2024) → 25.2% (o3, Tier 1-4) → 51.7% Tier1-3 / 35.4% Tier4 (GPT-5.5, 2026). Grok 4 and DeepSeek v3.2 2%, Llama 4 Maverick 1% (2026), showing steep difficulty cliff.
• Mathlib growth: >210k theorems, 100k definitions (May 2025); PhysLib 180k+ LOC (June 2026).
• LeanPhysBench: +11.9% average improvement when PhysLib provided in context.
• FormulaGate: 100% pass-through on MMLU (1,350), MMLU-Pro (200), GPQA Diamond (200) as physics instrument staying silent on general text—demonstrating selective verification.

8. Architectural Blueprint for Math-Physics ASI

Based on synthesis, a viable ASI limited to math/physics requires:

1. Ingestion Layer: LLM-based auto-formalizer converting textbooks, papers, competition problems to Lean 4 statements (80M synthetic statements pattern from AlphaProof).
2. Objective Core: Mathlib + PhysLib + SciLean as ground truth; knowledge graph (59+ nodes, 115+ typed edges) with evidence labels and Lean target audit.
3. Reasoning Loop: Policy network proposes tactics → Lean kernel evaluates proof state → Value network + MCTS search; CAS/Z3 gate validates numeric/symbolic steps.
4. Verification Gate: Binary decision—proof checks or not; if not, withhold or reroute. Implements explicit confidence targets and abstention, fixing evaluation bias that rewards guessing.
5. Learning Flywheel: Expert iteration—verified proofs reinforce training (AlphaProof flywheel); errors become negative examples.
6. Interface: Natural language explanation layer that cites formal artifacts, providing traceability (URI of theorem, not just fluent text).

This architecture treats LLMs as soft reasoners and formal systems as hard verifiers. It aligns with the 'physics narrow-domain superintelligence' definition: exceed general LLM capability in verified 3D/physics data while structurally bounded against hallucination.

9. Conclusion

LLMs cannot provide an objective knowledge base for mathematics and physics on their own. Their probabilistic objective, opacity, lack of provenance, and mathematically proven propensity to hallucinate make them unsuitable as sources of truth, especially where truth is mechanically checkable.

ASI limited to math and physics therefore requires specialized formats: formal proof assistants with large libraries (Lean 4 + mathlib + PhysLib), deterministic symbolic engines, explicit knowledge graphs with evidence labeling, and neuro-symbolic loops where LLMs propose and verifiers dispose. Empirical results from 2024-2026—from FrontierMath's <2% to 50%+ progression, from AlphaProof's silver-medal Lean proofs, from Pythos's 0% incorrect delivery, and from LeanPhysBench's +11.9% with formal context—converge on the same architectural lesson.

The productive framing is not LLM versus formal, but LLM inside formal: an LLM without Lean is a storyteller; Lean without an LLM is a library without a librarian. A math-physics ASI needs both, but objectivity must reside in the specialized, verifiable layer.

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